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1  # Note 06: Regression Models (LR / PL / MLR)
2  # =====
3
4  # Import Libraries
5  from sklearn.linear_model import LinearRegression
6  from sklearn.model_selection import train_test_split
7  from sklearn.metrics import mean_squared_error, r2_score
8  import numpy as np
9  import pandas as pd
10 from sklearn.preprocessing import PolynomialFeatures
11 from sklearn.pipeline import make_pipeline
12 from sklearn.model_selection import cross_val_score
13
14 # 01. Linear Regression (Order 01)
15 # =====
16
17 # Load Dataset
18 df = pd.read_csv("test_data.csv")
19
20 # Define X and Y Variables
21 X = df[['temperature']]
22 y = df['volt']
23
24 # Train - Test Split
25 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=
42)
26
27 # Define a Model
28 model = LinearRegression()
29 model.fit(X_train, y_train)
30 y_pred = model.predict(X_test)
31
32 # Calculate Values
33 rmse = np.sqrt(mean_squared_error(y_test, y_pred))
34 r2 = r2_score(y_test, y_pred)
35
36 # Print Values
37 print("\nOrder 01")
38 print("RMSE:", rmse)
39 print("R² : ", r2)
40
41 # 02. Polynomial Regression Model (Order > 1)
42 # =====
43
44 # Order
45 degree = 3
46
47 # Mutiple Models into One Model
48 model = make_pipeline(PolynomialFeatures(degree=degree), LinearRegression())
49 model.fit(X_train, y_train)
50 y_pred = model.predict(X_test)
51
52 # Calculate Values
53 rmse = np.sqrt(mean_squared_error(y_test, y_pred))
54 r2 = r2_score(y_test, y_pred)
55
56 # Print Values
57 print("\nOrder 03")
58 print("RMSE:", rmse)
59 print("R²:", r2)
60
61 # 03. Cross - Validation to Evaluate Best Model
62 # =====
63
64 # Test polynomial orders
65 orders = [1, 2, 3, 4, 5]
66
67 # Calculate RMSE Values
68 for order in orders:
69     model = make_pipeline(PolynomialFeatures(degree=order), LinearRegression())
70     # 5-fold cross-validation
71     scores = cross_val_score(model, X, y, cv=5, scoring='neg_root_mean_squared_error')

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72     # Convert negative RMSE to positive RMSE
73     rmse = -scores
74     # Print Results
75     print("\nOrder:", order)
76     print("RMSE for each fold:", rmse)
77     print("Average RMSE:", rmse.mean())
78     print()
79
80 # 04. Multiple Linear Regression
81 # =====
82
83 # Load Dataset
84 df = pd.read_csv("test_data.csv")
85
86 # Define X and y
87 X = df['temperature'].to_numpy().reshape(-1, 1)
88 # Feed Here More X Columns and It will Act as a MLR Problem
89 y = df['volt']
90 # -1 => Calculate Row Count Automatically / 1 => One Column
91
92 # Train-Test Split
93 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=
42)
94
95 # Define Model
96 model = LinearRegression()
97
98 # Train Model
99 model.fit(X_train, y_train)
100
101 # Prediction
102 y_pred = model.predict(X_test)
103
104 # Evaluation
105 rmse = np.sqrt(mean_squared_error(y_test, y_pred))
106 r2 = r2_score(y_test, y_pred)
107
108 # Print Results
109 print("RMSE:", rmse)
110 print("R²   :", r2)
111
112 # Coefficients
113 print("Intercept:", model.intercept_)
114 print("Coefficients:", model.coef_[0])

```